INTERVIEW

"We now want to be drivers of the bus"

Ashwini Vaishnaw, Union Minister of Electronics and Information Technology, spoke to Group Editorial Director Raj Chengappa about the Modi government's vision for and execution of the massive Rs 76,000 crore India Semiconductor Mission (ISM). Excerpts



Photograph by BANDEEP SINGH

Q. In 1984, India seemed ahead in the semiconductor race, having set up the Semi-Conductor Laboratory in Mohali. But 40 years later, we seem to have missed the bus—we are importing 95 per cent of our chips. What went wrong?

There is history behind that. It has been a period of missed opportunities, but I do not want to get into a political discussion on why. What is important, as our prime minister said, is that we should not be looking at the bus we missed, but must now become the drivers of this bus. And he has a very clear vision and laser-sharp focus on executing it.

Q. How different is the Indian Semiconductor Mission launched by the Modi government from previous such initiatives?

It is important to understand the nature of the industry. It works well when the entire ecosystem gets developed. More than 500 chemicals and over 50 gases are used in manufacturing chips. These are chemicals and gases where you measure purity not in parts per million, but per billion and trillion, as these are ultra, ultra pure materials. To get them, it is essential to ensure the stability of the entire grid and develop the complete ecosystem, so that the foundation of the industry is laid properly and it can run for the next 40 years. For that, we have to look at the entire value chain, from design, fabrication and assembly testing to marking and packaging supported by a series of EDA or electronic design automation tools.

Q. Will ISM help develop this ecosystem?

We have worked diligently with every ecosystem player, whether it is those who make chemical gases, equipment manufacturers and suppliers along with the state governments, the local level panchayat and municipal governments. This industry requires a minimum of 10 years to get established. China took, I think, about 20-plus years. Today, we are at a stage where 10 units are under construction or under planning. In the value chain design, we all know we have very good indigenous strength. So, it is really satisfying to know that we shall soon have entire end-to-end chip-making.

"This industry works well only **when the entire ecosystem is developed**, from design, fabrication, assembly testing to packaging. That takes a minimum of 10 years"